

For motors rated above 4000 kW, add 3 dB to all levels in Table 11.25. For motors rated between 300 and 750 kW, subtract 3 dB from all levels.

Table 11.25 Sound power levels of large electric motors^a

Octave band centre frequency (Hz)	1800 and 3600 rpm	1200 rpm	900 rpm	720 rpm and lower	250 and 400 rpm vertical
315	94	88	88	88	86
63	96	90	90	90	87
125	98	92	92	92	88
250	98	93	93	93	88
500	98	93	93	93	88
1000	98	93	96	98	98
2000	98	98	96	92	88
4000	95	88	88	83	78
8000	88	81	81	75	68

^a Applies to induction motors rated between 750 and 4000 kW; includes drip-proof and P-1 and WP-2 enclosures (with no acoustical specification by the customer). Data from Edison Electric Institute (1978).

11.15 GENERATORS

The overall sound power levels radiated by generators (excluding the driver) can be calculated using the following equation:

$$L_w = 84 + 10 \log_{10} MW + 6.6 \log_{10} RPM \quad (\text{dB re } 10^{-12} \text{ W}) \qquad (11.93)$$

To obtain the octave band levels, the values in Table 11.26 should be subtracted from the overall level calculated using Equation (11.93).

11.16 TRANSFORMERS

The sound power radiated in octave bands by a transformer is related to its NEMA (National Electrical Manufacturers Association, 1980) sound level rating and the total surface area of its four side walls by:

$$L_w = N_R + 10 \log_{10} S + C \quad (\text{dB re } 10^{-12} \text{ W}) \qquad (11.94)$$

The quantity N_R is the NEMA sound level rating, which is the average sound pressure level measured around the transformer at a distance of 0.35 m, and is generally specified by the transformer manufacturer. The quantity, S is the surface area (m²) of the four transformer walls (excluding the top). Values of the correction term C are listed in Table 11.27.

Table 11.26 Octave band corrections for generator noise

Octave band centre frequency (Hz)	Value to be subtracted from overall sound power level L_w (dB)
31.5	11
63	8
125	7
250	7
500	7
1000	9
2000	11
4000	14
8000	19

Data from Army, Air Force and Navy, USA (1983a).

Table 11.27 Values of correction C of Equation (11.94) for transformer noise

Octave band centre frequency (Hz)	Octave band corrections (dB)		
	Location 1 ^a	Location 2 ^b	Location 3 ^c
31.5	-1	-1	-1
63	5	8	8
125	7	13	13
250	2	8	12
500	2	8	12
1000	-4	-1	6
2000	-9	-9	1
4000	-14	-14	-4
8000	-21	-21	-11

^a Outdoors, or indoors in a large mechanical room with a large amount of mechanical equipment.

^b Indoors in small rooms, or large rooms with only a small amount of other equipment.

^c Any critical location where a problem would result if the transformer should become noisy above its NEMA rating, following installation.

Data from Army, Air Force and Navy, USA (1983a).

Values of C are lower than actual for air-filled transformers and higher than actual for oil-filled transformers. Consult manufacturer's data for values of sound level rating. Typical values are listed in Table 11.28.

Table 11.28 Transformer sound level ratings, N_R for transformers built after 1995

Transformer power (kVA)	Sound level (dB)		
	Standard		Super quiet core
	No fan	Fan	
100-300	55	67	52
301-500	56	67	52
501-700	57	67	53
701-1000	58	67	56
1001-1500	60	67	58
1501-2000	61	67	60
2001-2500	62	67	62
2501-3000	63	67	63
3001-4000	64	67	63
4001-5000	65	67	64
5001-6000	66	68	65
6001-7500	67	69	66
7501-10000	68	70	67
10001-12000	69	71	68
12001-15000	70	72	69
15001-20000	71	73	70

11.17 GEARS

The following equation gives octave band sound pressure levels for gearboxes in all frequency bands at and above 125 Hz, at a distance of 1 m from the gearbox (Army, Air Force and Navy, USA, 1983a):

$$L_p = 78 + 4 \log_{10} kW + 3 \log_{10} RPM \quad (\text{dB re } 10^{-12} \text{ W}) \tag{11.95}$$

where kW is the power transmitted by the gearbox, and RPM is the rotational speed of the slowest gear shaft. For the 63 Hz octave band, subtract 3 dB, and for the 31 Hz band subtract 6 dB from the value calculated using Equation (11.95).

These noise levels are applicable to spur gears, and may be reduced somewhat (by up to 30 dB) by replacing the spur gear with a quieter helical or herringbone design. The actual noise reduction (compared to a straight spur gear) is given very approximately by $13 + 20 \log_{10} Q_a$, where Q_a is the number of teeth that would be intersected by a straight line parallel to the gear shaft. For double helical or herringbone gears, the number of intersected teeth would only be for one helix, not both.